

HW0414

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \epsilon_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- a. We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(\epsilon_i | x_i, FEMALE = 0) = \sigma_u^2$ and $\text{var}(\epsilon_i | x_i, FEMALE = 1) = \sigma_v^2$. We specifically wish to test the null hypothesis $\sigma_u^2 = \sigma_v^2$ against $\sigma_u^2 \neq \sigma_v^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_u = 97161.9174$. The regression using data on females yields $\hat{\sigma}_v^2 = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(\epsilon_i | x_i, MARRIED = 0) = \sigma_{\text{SINGLE}}^2$ and $\text{var}(\epsilon_i | x_i, MARRIED = 1) = \sigma_{\text{MARRIED}}^2$. Specify the null hypothesis $\sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$ versus the alternative hypothesis $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE_i + \epsilon_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- c. Following the regression in part (b), we carry out the NF^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?
- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\begin{array}{l} WAGE = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE \\ (se) \quad (2.36) \quad (0.14) \quad (0.031) \quad (1.05) \quad (0.81) \\ (robse) \quad (2.50) \quad (0.16) \quad (0.029) \quad (0.84) \quad (0.80) \end{array}$$

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- f. If we add $MARRIED$ to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

$$a. df_M = 577 - 3 - 1 = 573$$

$$\hat{\sigma}_M^2 = \frac{SSE_M}{df_M} = \frac{97161.9174}{573} = 169.569$$

$$F\text{-statistic} = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} = \frac{169.569}{12.024} = 1.193$$

$$df_M = 1000 - 577 - 3 - 1 = 419$$

在 5% 的 significant level, critical value:

$$F_{(1-0.05, 573, 419)} = 1.1968, F_{(0.05, 573, 419)} = 0.8377$$

$\Rightarrow$ reject region: $F\text{-statistic} > 1.1968$ or $F\text{-statistic} < 0.8377$

$\therefore F\text{-statistic} = 1.193$ doesn't fall in the reject region $\Rightarrow H_0$ is not rejected.

$\Rightarrow$ There's not enough evidence to say that $\sigma_M^2 \neq \sigma_F^2$.

$$b. F\text{-statistic} = \frac{\hat{\sigma}_{\text{MARRIED}}^2}{\hat{\sigma}_{\text{SINGLE}}^2} = \frac{\frac{100703.0471}{600}}{\frac{56231.0382}{395}} = 1.189$$

$\Rightarrow$ reject region: $F\text{-statistic} > 1.1649$

$\therefore F\text{-statistic} = 1.189$ falls in the reject region

$\Rightarrow H_0$ is rejected.

$\Rightarrow$ There's not enough evidence to say that $\sigma_M^2 = \sigma_F^2$.

C.

$$H_0: \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = \alpha_5 = 0.$$

$$H_1: H_0 \text{ is not true.}$$

$$\chi^2_{1-0.05, 4} = 9.4877$$

$$\text{test statistic: } \chi^2 = 59.03$$

$$\therefore \chi^2 = 59.03 > \chi^2_{1-0.05, 4} = 9.4877$$

$\Rightarrow H_0$ is rejected.

$\Rightarrow$ There's a difference between the error variation,

but it didn't point out which of them is different.

d.

原變數: EDUC, EXPER, METRO, FEMALE $\Rightarrow$ 共 4 項.

平方項: EDUC², EXPER², METRO², FEMALE² $\Rightarrow$ 共 4 項.

交乘項: $3 \times 2 = 6$.

$$\Rightarrow df = 4 + 4 + 6 = 14$$

H_0 : Homoskedasticity v.s. H_1 : Heteroskedasticity

$$\chi^2_{1-0.05, 14} = 23.685$$

$$\text{test statistic: } \chi^2 = 78.82$$

$$\therefore \chi^2 = 78.82 > \chi^2_{1-0.05, 14} = 23.685$$

$\Rightarrow H_0$ is rejected.

$\Rightarrow$ There's significant evidence of heteroskedasticity.

e.

C.I. for EXPER, METRO, FEMALE gotten narrower,

C.I. for EDUC gotten wider.

$\Rightarrow$ 受 heteroskedasticity 影響.

f.

There is no conflict.

$\therefore$ (b) 在檢驗是否為同質變異,

(c) 則在檢驗是否係數是否顯著 $\neq 0$.

8.8.2 Computer Exercises

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_0 + \beta_1 INCOME + \beta_2 AGE + \beta_3 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld-Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_e^2 = \sigma^2 INCOME^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

a.

```
#a
> #a
> model <- lm(miles ~ income + age + kids, data = vacation)
> summary(model)

Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:
    Min       1Q   Median       3Q      Max
-1198.14   -295.31    17.98   287.54  1549.41

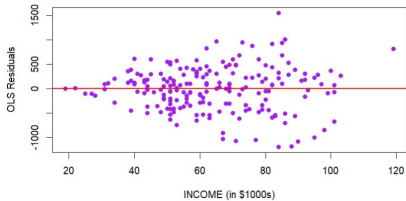
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -391.548    169.775   -2.306  0.0221 **
income         14.201      1.800    7.889 2.10e-13 ***
age           15.741      3.757    4.189 4.23e-05 ***
kids          -81.826     27.130   -3.016  0.0029 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406, Adjusted R-squared:  0.3305
F-statistic: 33.75 on 3 and 196 DF, p-value: < 2.2e-16

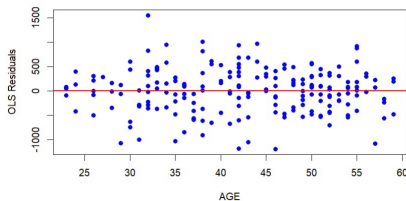
> confint(model, "kids", level = 0.95)
           2.5 %    97.5 %
kids -135.3298 -28.32302
```

b.

OLS Residuals vs INCOME → Heteroskedasticity



OLS Residuals vs AGE → Homoskedasticity



c. H_0 : Homoskedasticity v.s. H_1 : Heteroskedasticity

```
> #income小組大
> vacation$order <- vacation$order[order(vacation$income), ]
>
> #把中間20筆拿掉，保留first90和last90
> group1 <- vacation$order[1:90, ]
> group2 <- vacation$order[111:200, ] #共200筆，所以從90起是從111開始
>
> #分別估計OLS模型
> model1 <- lm(miles ~ income+age+kids, data=group1)
> model2 <- lm(miles ~ income+age+kids, data=group2)
>
> #RSS
> RSS1 <- sum(resid(model1)^2)
> RSS2 <- sum(resid(model2)^2)
>
> #統計量
> F_stat <- RSS2 / RSS1
>
> #比對臨界值或計算p-value
> df1 <- model1$df.residual #總組df
> df2 <- model2$df.residual #總組df
> p_value <- 1 - pf(F_stat, df1, df2)
>
> cat("F-statistic = ", F_stat, "\np-value = ", p_value, "\n")
F-statistic = 3.104061
p-value = 1.64001e-07
```

F-statistic = 3.104061 > Critical value = 1.4286.

⇒ H_0 is rejected

⇒ There is heteroskedasticity in the model's errors.

d.

```
> model <- lm(miles ~ income+age+kids, data=vacation)
>
> #robust標準誤差
> robust_se <- vcovHC(model, type = "HC1")
> coeftest(model, vcov=robust_se)

t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -391.5480    162.6548   -2.447 0.0066190 **
income         14.2013      1.9389    7.3246 6.083e-12 ***
age           15.7409      3.9657    3.9692 0.0001011 ***
kids          -81.8264    29.1544  -2.8067 0.0051112 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

>
> #KIDS的95%信賴區間
> confint_robust <- coeftest(model, vcov = robust_se, level=0.95)
> confint_robust["kids", ]
           2.5 %    97.5 %
kids -139.32297 -24.32986
```

e.

```
> #權重 = w.i-1 / income.i
> weights <- 1 / vacation$income
> #使用WLS估計模型，帶入GLS
> gls_model <- lm(miles ~ income + age + kids, data = vacation, weights = weights)
> summary(gls_model)

Call:
lm(formula = miles ~ income + age + kids, data = vacation, weights = weights)

Weighted Residuals:
    Min       1Q   Median       3Q      Max
-131.336   -37.983    2.073   35.993   169.751

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -408.374    145.717   -2.803 0.00558 **
income         13.971      1.648    8.476 5.54e-15 ***
age           16.348      4.422    3.696 0.000348 ***
kids          -78.363    24.736   -3.168 0.00178 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 54.43 on 196 degrees of freedom
Multiple R-squared:  0.3907, Adjusted R-squared:  0.3814
F-statistic: 41.9 on 3 and 196 DF, p-value: < 2.2e-16

> confint_conventional <- confint(gls_model)[["kids", 1]]
> robust_se_gls <- vcovHC(gls_model, type = "HC1")
> confint_robust <- coeftest(gls_model, vcov = robust_se_gls, level = 0.95)[["kids", 1]]
> cat("Conventional GLS C.I. (kids): ", "\n")
Conventional GLS C.I. (kids):
< print(confint_conventional)
           2.5 %    97.5 %
kids -127.1433 -29.5814
> cat("Robust GLS C.I. (kids): ", "\n")
Robust GLS C.I. (kids):
           2.5 %    97.5 %
kids -127.71734 -29.00934
```

⇒ GLS 點估計的區間較小，且在左移。